

UNIVERSITY OF CAPE TOWN

DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Parametric surfaces I

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0.1 Definitions and properties

Now that we have dealt with scalar valued functions of several variables, we are in a position to present vector functions of several variables. In analogy with vector functions of a single variable which are used in the description of curves, we use vector functions of two variables in order to describe those many surfaces in the space which are not graphs of functions of two variables.

So far, we studied particular surfaces $\mathcal{S}$ in the space $\mathbb{R}^3$ which are graphs of functions $f : A \subset \mathbb{R}^2 \rightarrow \mathbb{R}$. That is,

$$\mathcal{S} := \{(x, y, z) \in A \times \mathbb{R} : z = f(x, y)\} \subset \mathbb{R}^3.$$

Such a surface $\mathcal{S}$ is described by the vector function $\vec{r} : A \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(x, y) := (x, y, f(x, y)) \quad \forall (x, y) \in A.$$

Definition 0.1 Let $\vec{r} : D \rightarrow \mathbb{R}^3$ be a vector function. Then the image $\mathcal{S} = \vec{r}(D)$ is called a **parametric surface** and the vector function $\vec{r}$ is called **parametric representation of $\mathcal{S}$** . Each point $(x, y, z) \in \mathcal{S}$ is described through the function $\vec{r}$ as

$$x = r_1(s, t), \quad y = r_2(s, t), \quad z = r_3(s, t) \quad \text{for some } (s, t) \in D. \quad (1)$$

The equations (1) are called the **parametric equations** of the surface $\mathcal{S}$.

Sometimes one is able to eliminate the variables s, t from the parametric equations (1) and get an equation called the **cartesian equation** of the surface $\mathcal{S}$.

In these lectures, we will be using the terminology "**Let $\mathcal{S}$ be the surface described by the vector function $\vec{r} : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$** ".

Example 0.2

- (a) The lateral surface $\mathcal{S}$ of a circular cylinder of height $h > 0$ and radius $r > 0$ is described by the vector function:

$$\vec{r}(s, t) := (r \cos s, r \sin s, th) \quad \text{with } (s, t) \in D = [0, 2\pi] \times [0, 1].$$

- (b) The surface of the sphere centered at the point (a, b, c) with radius $R > 0$ is described by the vector function

$$\vec{r}(s, t) := (a + R \sin t \cos s, b + R \sin t \sin s, c + R \cos t) \quad \forall (s, t) \in [0, 2\pi] \times [0, \pi].$$

We will be dealing with vector functions of the form

$$\vec{r}: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$$

where $D = [a, b] \times [c, d]$ or (in general D is a subset of $\mathbb{R}^2$ with a connected interior). Some authors would prefer the terminology: “ D is a continuous deformation of a rectangle”. For the moment there is no need to bother too much about the meanings of “connected interior” or “continuous deformation”.

Definition 0.3 (Regular surface)

A surface $\mathcal{S}$ described by the vector function $\vec{r}: D = [a, b] \times [c, d] \rightarrow \mathbb{R}^3$ is said to be **regular** if

- (i) the partial derivatives $\vec{r}_s$ and $\vec{r}_t$ are continuous;
- (ii) the cross-product $\vec{r}_s(s, t) \times \vec{r}_t(s, t) \neq (0, 0, 0)$ for every $(s, t) \in (a, b) \times (c, d)$.

The condition (ii) is often formulated as:

(ii)' the matrix

$$D\vec{r}(s, t) = \frac{\partial(r_1, r_2, r_3)}{\partial(s, t)}(s, t) = \begin{pmatrix} \frac{\partial r_1}{\partial s}(s, t) & \frac{\partial r_1}{\partial t}(s, t) \\ \frac{\partial r_2}{\partial s}(s, t) & \frac{\partial r_2}{\partial t}(s, t) \\ \frac{\partial r_3}{\partial s}(s, t) & \frac{\partial r_3}{\partial t}(s, t) \end{pmatrix}$$

called the **Jacobian matrix** of the vector function $\vec{r}$ has **rank two** using a terminology from Linear Algebra.

Example 0.4

- (a) Let $f: D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function with continuous partial derivatives f_x and f_y . Then the graph $\mathcal{S}$ of the function f is a regular surface. In fact, as mentioned before, the graph $\mathcal{S}$ is described by the vector function

$$\vec{r}(x, y) := (x, y, f(x, y)) \quad \forall (x, y) \in D.$$

The partial derivatives

$$\vec{r}_x(x, y) = (1, 0, f_x(x, y)) \quad \text{and} \quad \vec{r}_y(x, y) = (0, 1, f_y(x, y))$$

are continuous. Also,

$$\vec{r}_x(x, y) \times \vec{r}_y(x, y) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & f_x(x, y) \\ 0 & 1 & f_y(x, y) \end{vmatrix} = (-f_x(x, y), -f_y(x, y), 1) \neq (0, 0, 0).$$

- (b) Check that the lateral surface $\mathcal{S}$ of the cylinder described in Example 0.2 is a regular surface.

Remark 0.5 Notice that the notion of regularity for a surface depends on the parametric representation used. In fact, consider the surface $\mathcal{S}$ of the circular upward cone with vertex at the origin and height $h > 0$.

- (a) On the one hand, $\mathcal{S}$ can be described as the graph of the function $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) := h\sqrt{x^2 + y^2} \quad \forall (x, y) \in D := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}.$$

Therefore, $\mathcal{S}$ is described by the vector function $\vec{r}(x, y) = (x, y, f(x, y))$ which does not have partial derivatives at $(0, 0)$. Hence, the surface is not regular when described as a graph.

- (b) On the other hand, the surface $\mathcal{S}$ can also be described by the vector function $\vec{r} : [0, 1] \times [0, 2\pi] \rightarrow \mathbb{R}^3$ defined by

$$\vec{r}(s, t) := (s \cos t, s \sin t, sh) \quad \forall (s, t) \in [0, 1] \times [0, 2\pi]$$

for which we notice that

$$\vec{r}_s(s, t) = (\cos t, \sin t, h) \quad \text{and} \quad \vec{r}_t(s, t) = (-s \sin t, s \cos t, 0)$$

which are continuous and

$$\vec{r}_s(s, t) \times \vec{r}_t(s, t) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos t & \sin t & h \\ -s \sin t & s \cos t & 0 \end{vmatrix} = (-hs \cos t, -hs \sin t, s) \neq (0, 0, 0).$$

for every $(s, t) \in (0, 1) \times (0, 2\pi)$ (interior of the domain). Therefore, the surface $\mathcal{S}$ is regular through this description.

0.2 Cylindrical and spherical coordinates

0.2.1 Cylindrical coordinates:

For the description of the cylindrical coordinates, we take a generic point $M(x, y, z)$ that we project orthogonally onto the xy -plane to obtain the point $A(x, y, 0)$. Now, we describe (x, y) in polar coordinates (r, θ) where r is the distance between the point A and the origin O while θ is the angle between the segment line OA and the positive x -axis, by

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta.$$

We then obtain the set of coordinates (r, θ, z) defined through the equations:

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z.$$

0.2.2 Spherical coordinates:

For the description of the spherical coordinates, we take a generic point $M(x, y, z)$ and consider the distance ρ between the point M and the origin O ($\rho := \sqrt{x^2 + y^2 + z^2}$) and the angle ϕ between the segment line OM and the positive z -axis. Now, if we project the point M orthogonal onto the z -axis, we get $z = \rho \cos \phi$. On the other hand, if we project orthogonally the point M onto the xy -plane we obtain the point $A(x, y, 0)$. Now, we describe (x, y) in polar coordinates as

above with the difference that we use the triangle OMA in order to express r in terms of ρ and ϕ and get that $r = \rho \sin \phi$. Therefore, we obtain the set of coordinates (ρ, θ, ϕ) defined through the equations:

$$x = \rho \cos \theta \sin \phi, \quad y = \rho \sin \theta \sin \phi, \quad z = \rho \cos \phi.$$

Exercises 0.6

1. Convert the coordinates as indicated:

(a) $(3, \frac{\pi}{3}, -4)$ from cylindrical to cartesian coordinates;

In cylindrical coordinates $r = 3$, $\theta = \frac{\pi}{3}$ and $z = -4$. Now we find

$$x = r \cos \theta = 3 \cos(\pi/3) = \frac{3}{2}, \quad y = r \sin \theta = 3 \sin(\pi/3) = \frac{3\sqrt{3}}{2}.$$

So, we get the point $(\frac{3}{2}, \frac{3\sqrt{3}}{2}, -4)$.

(b) $(-2, 2, 3)$ from cartesian to cylindrical coordinates;

We have $x = -2$, $y = 2$ and $z = 3$. We seek

$$r = \sqrt{x^2 + y^2} = 2\sqrt{2}, \quad \cos \theta = \frac{-2}{2\sqrt{2}} = -\frac{1}{\sqrt{2}}, \quad \sin \theta = \frac{2}{2\sqrt{2}} = \frac{1}{\sqrt{2}} \Rightarrow \theta = \frac{3\pi}{4}.$$

So, we obtain the cylindrical coordinates $(2\sqrt{2}, \frac{3\pi}{4}, 3)$.

(c) $(8, \frac{\pi}{4}, \frac{\pi}{6})$ from spherical to cartesian coordinates.

Spherical coordinates:

$$\rho = 8, \quad \theta = \frac{\pi}{4}, \quad \phi = \frac{\pi}{6} \quad \text{with } x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi.$$

So, find x, y, z .

2. Make the required change in the given equation:

(i) $x^2 - y^2 = 16$ to cylindrical coordinates;

(ii) $x^2 + y^2 - z^2 = 1$ to spherical coordinates;

(iii) $\rho = 2 \cos \phi$ to cylindrical coordinates.

3. Convert into (i) cylindrical equation and (ii) spherical equation.

(a) $z = 2x^2 + 2y^2$;

(b) $x^2 + y^2 - 2z^2 = 3$.

The next set of notes will be on the notion of tangent plane to a parametric surface!